

PFC - SMART BOXING GLOVE

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Course: LEETC

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INTRODUCTION

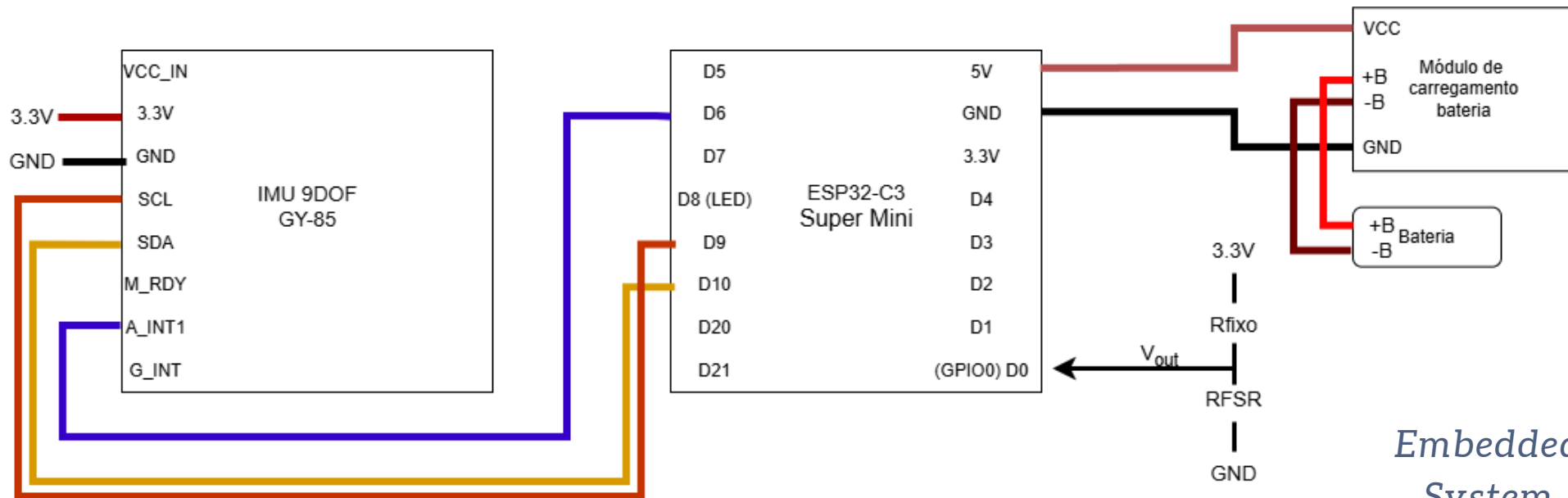
Boxing coaches still lack access to precise quantitative measurements, relying primarily on subjective visual observation of athletes performance.

This project aims to unite sport and technology through the **Smart Boxing Glove** that can help athletes track their performance.

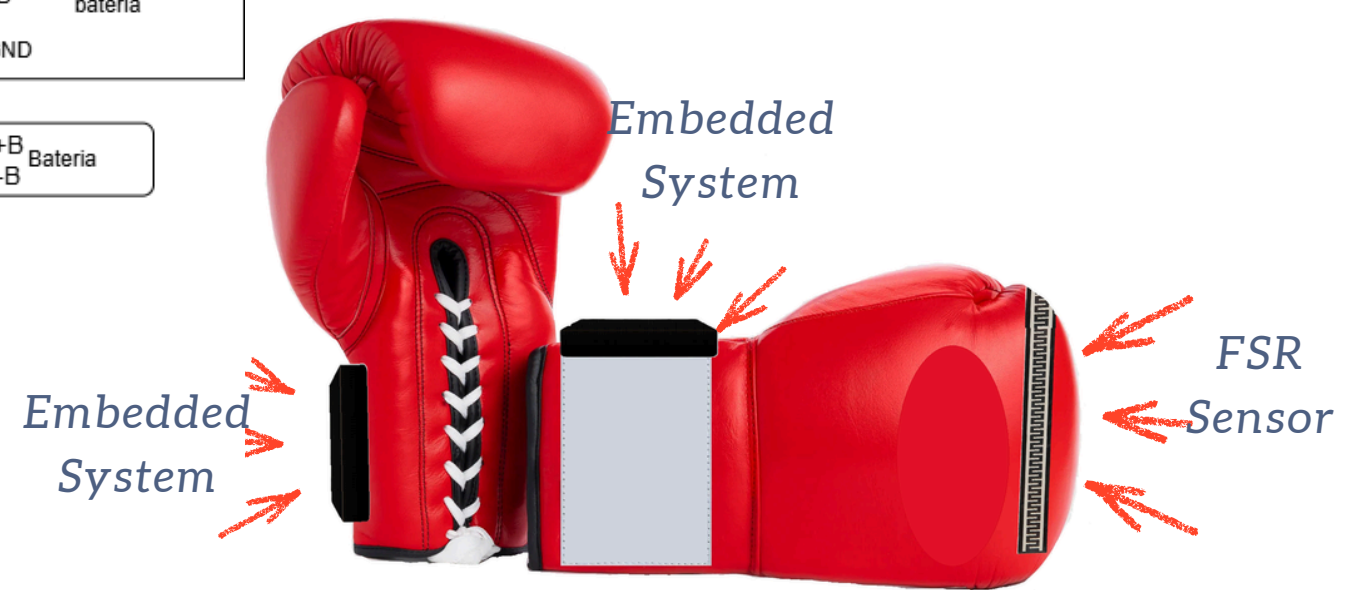
MAIN COMPONENTS

- Microcontroller: ESP32-C3 Supermini
- FSR Sensors
- IMU 9DOF GY-85

HARDWARE



- I2C communication
- FSR sensor voltage divider
- ADXL345 interrupt
- Battery with 231 mAH



- Protective capsule in low impact area
- FSR sensor in the area of greatest impact

SOFTWARE



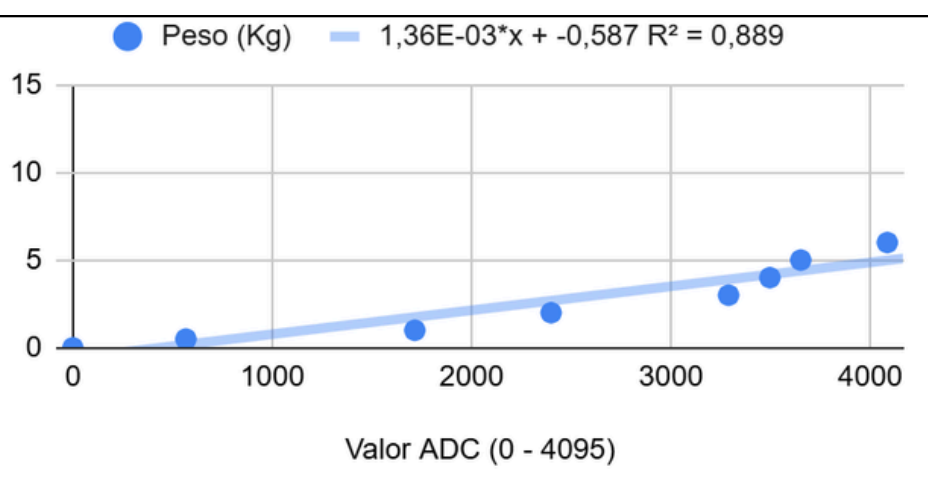
- Communication
- RTOS
- Connection Management
- Processing
- Visualization
- Operating Modes.

APP METRICS

- STRIKE FORCE
- STRIKE SPEED
- 3D TRAJECTORY

EXPERIMENTAL TESTS AND RESULTS

FSR Sensor Characterization



BLE Data Integrity



Strike Execution Analysis

Tipo de golpe	Competição	Manutenção	Média (segundos)
Jab	0,275	0,268	0,271
Direto	0,258	0,304	0,281
Cross	0,383	0,446	0,415
Uppercut	0,356	0,419	0,387

App Visualization

